

Summary- Housing Prices Dataset

Overall Info

1. The dataset had:
 - 545 rows/records
 - 13 columns - 12 features, 1 target (price).
2. The dataset had no column containing missing values.
3. Categorical values such as mainroad, guestroom, basement, hotwaterheating, airconditioning, prefaea, furnishingstatus were converted to numerical values using binary encoding and ordinal encoding (for furnishingstatus).
4. The dataset was split into train and test (80/20 split) as given.
5. 2 models were trained on it - Linear Regression & Random Forest Regressor.
6. Linear Regression had better R^2 Score.
7. Although hotwaterheating showed a weak correlation with house price, removing it slightly reduced model performance. Therefore, it was included again in the final dataset.

Which features influence house price the most?

Main features that contribute to the price are

1. area
2. bathrooms
3. airconditioning
4. stories
5. parking
6. bedrooms

How accurate was your model (in plain terms)?

- The Linear Regression model had 0.6542 R^2 Score, meaning it explained around 65% variation in the data.
- The Random Forest Regressor model has lower R^2 Score than Linear Regression model - 0.6148, meaning it explained around 61% variation in the data.
- In plain terms, the Linear Regression model has better accuracy than Random Forest Regressor and predicted the house prices fairly close to the actual value. However, 35% of

the variation couldn't be explained by existing features, leaving a major scope for improvement.

What surprised you in the data?

- It was surprising that the number of bathrooms had a stronger relationship with house prices than the number of bedrooms. I expected bedrooms to be among the top factors affecting price, but bathrooms showed a higher correlation with house prices.
- I initially dropped hotwaterheating as it had weak correlation with price and even other features. However, the R^2 Score increased by 1-2 % upon retaining it. So even the features with low correlation can sometimes prove to be meaningful.

One recommendation for a real estate business based on your findings

From my observations, features like area(obvious), bathrooms, airconditioning, stories, parking affect the prices even more than bedrooms. So, real estate businesses should focus on these features and their development, availability, quality/quantity significantly.